

Kinks, Breathers, and the Rest-Kink Spectrum in the Sine–Gordon Equation

HCS Research Program

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Abstract

We give a source-local theorem for the one-dimensional sine–Gordonequation $u_{tt} - u_{xx} + \sin u = 0$. The declared coherent families close exactly: finite-energy monotone travelling heteroclinics in every integer vacuum layer are the subluminal Lorentz kink and antikink, while the zero-charge breathers have an explicit rest frequency and a Lorentz energy–momentum ledger. At the rest kink the Hessian is the factored Pöschl–Teller operator with translation kernel and essential edge one. The period $2\pi/\Omega$ is a comoving breather clock, not a fixed-laboratory-point period after a nonzero boost. This is not a classification of every finite-energy solution and it is not an arithmetic or Hilbert–Pólya construction.

1 Frozen model and travelling heteroclinics

We use the potential $V(u) = 1 - \cos u$, with vacua $u = 2\pi k$, and define

$$E[u] = \int_{\mathbb{R}} \left[\frac{1}{2}(u_t^2 + u_x^2) + 1 - \cos u \right] dx, \quad P[u] = - \int_{\mathbb{R}} u_t u_x dx. \quad (1)$$

The minus sign in P is fixed here and in the certificate. We begin with the unrestricted ansatz $u(x, t) = U(x - vt)$, whose profile equation is $(v^2 - 1)U'' + \sin U = 0$; the speed inequality in the theorem justifies the Lorentz coordinate $\xi = \gamma_v(x - vt - x_0)$.

Theorem 1 (kink closure). *Every nonconstant finite-energy monotone travelling heteroclinic in the declared family has an integer vacuum layer $k \in \mathbb{Z}$ and, up to translation, is*

$$U_{k,+}(\xi) = 2\pi k + 4 \arctan(e^\xi), \quad U_{k,-}(\xi) = 2\pi k + 4 \arctan(e^{-\xi}), \quad (2)$$

and necessarily $|v| < 1$. Their charges are $Q = +1$ and $Q = -1$, and

$$E = 8\gamma_v, \quad P = 8\gamma_v v, \quad E^2 - P^2 = 64. \quad (3)$$

Proof. Multiplying the unrestricted profile equation $(v^2 - 1)U'' + \sin U = 0$ by U' and using the vacuum limits gives

$$\frac{1}{2}(v^2 - 1)(U')^2 = \cos U - 1.$$

The right side is nonpositive and a nonconstant heteroclinic has $U' \neq 0$ on an open set, hence $v^2 - 1 < 0$ and $|v| < 1$. In the resulting Lorentz coordinate, the first integral is $\frac{1}{2}(U')^2 = 1 - \cos U$. Monotonicity and the 2π -periodicity of the potential select the integer-layer profiles (2); the opposite sign is the antikink. In the canonical $k = 0$ rest frame $U'_+ = 2 \operatorname{sech} \xi$ and $\int 4 \operatorname{sech}^2 \xi d\xi = 8$. Lorentz change of variables gives (3) with the convention (1). $\square$

2 Breathers and the Lorentz ledger

For $0 < \Omega < 1$ let $\eta = (1 - \Omega^2)^{1/2}$. The exact rest breather is

$$u_B(x, t) = 4 \arctan \left(\frac{\eta \sin(\Omega t)}{\Omega \cosh(\eta x)} \right). \quad (4)$$

It has $Q = 0$, rest momentum zero, rest energy $E_0 = 16\eta$, and internal period $T_0 = 2\pi/\Omega$. For $|V| < 1$, set

$$\xi = \gamma_V(x - Vt), \quad \tau = \gamma_V(t - Vx), \quad \gamma_V = (1 - V^2)^{-1/2}, \quad (5)$$

and replace (x, t) in (4) by (ξ, τ) . Then

$$E = 16\eta\gamma_V, \quad P = 16\eta\gamma_V V, \quad E^2 - P^2 = (16\eta)^2. \quad (6)$$

The period T_0 belongs to the rest/comoving proper clock. For $V \neq 0$ we make no claim of a strict period at a fixed laboratory x ; the dependence of τ on both t and x is essential. At $V = 0$ the rest statement is the usual fixed- x period.

For smooth finite-energy solutions, the stress-energy identity gives $\frac{d}{dt}E = 0$ and $\frac{d}{dt}P = 0$ (boundary fluxes vanish). Thus the two ledgers above are exact identities, not fitted numerical laws.

Table 1: Boundary faces retained by the certificate.

face	coherent object	limiting statement
$ v \uparrow 1$	kink	width collapses; $E, P \rightarrow \infty$
$\Omega \uparrow 1$	breather	zero-amplitude vacuum; $E_0 \rightarrow 0$
$\Omega \downarrow 0$	breather	infinite-period separatrix; $E_0 \rightarrow 16$
$V = 0 / V \neq 0$	breather	rest clock / comoving clock only

3 Rest-kink Hessian and spectral boundary

At $v = 0$, linearization about U_+ gives

$$L_K = -\partial_x^2 + \cos U_+(x) = -\partial_x^2 + 1 - 2 \operatorname{sech}^2 x. \quad (7)$$

With $A = \partial_x + \tanh x$ and $A^* = -\partial_x + \tanh x$,

$$L_K = A^* A, \quad L_K(2 \operatorname{sech} x) = 0. \quad (8)$$

Proposition 2 (rest-kink spectrum). *On $L^2(\mathbb{R})$ with the standard H^2 domain,*

$$\sigma(L_K) = \{0\} \cup [1, \infty), \quad \ker L_K = \operatorname{span}\{2 \operatorname{sech} x\}.$$

There is no internal discrete mode below the essential edge.

Proof. The factorization makes L_K nonnegative and the first-order equation $Af = 0$ gives the unique square-integrable zero mode. The potential tends to one at both ends, so Weyl comparison gives essential spectrum $[1, \infty)$. The one-dimensional Pöschl–Teller intertwining (equivalently, the Sturm oscillation count) leaves no additional bound state for this $2 \operatorname{sech}^2$ well. $\square$

4 Scope boundary and reproducibility

The statement above is a declared coherent-family theorem, not a classification of every finite-energy sine–Gordon field, and the rest-kink spectrum is not a boosted-breather nonlinear stability theorem. The breather frequency is a continuous parameter, so these solutions do not furnish an isolated primitive periodic-orbit owner. Accordingly the locked Route-A record is

`(A0_FAIL,A1_WEAK,A2_FAIL,A3_FAIL,A4_NATURAL_QUANTIZATION),`

with `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`.

The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`; no target primes or zeros, Euler factors, root numbers, automorphy, target divisors, or Hilbert–Pólya operator are used.

The receipt has 6 kink, 6 breather, 5 Hessian, 8 boundary, and 4 Lorentz rows. Its producer-independent checker (289 assertions), symbolic cross-check, byte replay, and 41/41 repaired-hash hostile mutations are part of the artifact. Two fresh fixed-epoch builds are required for each of the three revision rounds.

Source note

The classical fluxon context is recorded by McLaughlin and Scott [1]; all displayed formulas and limits are re-derived in this source-local artifact, and the citation is not a priority or completeness claim.

References

- [1] J. M. McLaughlin and A. C. Scott, “Perturbation analysis of fluxon dynamics,” *Physical Review A* 18(4), 1652–1680 (1978). DOI: 10.1103/PhysRevA.18.1652.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, primitive-orbit, or Hilbert–Pólya claim. **Data and code.** The exact profiles, ledgers, spectrum, independent checks, and build instructions are released with HCS-C236. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.